Temporal Causal Graph Retrieval Under Distribution Shift in Financial Markets

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1 Introduction

Financial event retrieval and intelligence systems have become increasingly important for analyzing rapidly evolving market conditions, enabling automated understanding of geopolitical events, supply-chain disruptions, commodity fluctuations, and macroeconomic shocks. Traditional financial retrieval approaches primarily rely on semantic similarity, vector embeddings, and correlation-based graph structures to identify relevant historical events. While these methods achieve strong retrieval performance under stable market conditions, they often fail during temporal market distribution shifts such as the COVID-19 pandemic, geopolitical conflicts, and inflation-driven regime transitions. Such failures occur because correlation-based systems capture co-occurrence patterns rather than underlying causal relationships governing event propagation in financial ecosystems.

Recent advances in graph-based financial intelligence and retrieval-augmented generation have demonstrated the effectiveness of temporal graphs and knowledge-aware retrieval for financial forecasting and event reasoning. However, most existing approaches model event relationships using static or undirected dependencies, limiting their ability to distinguish upstream causal events from downstream effects. In practical financial scenarios, causal direction plays a critical role. For example, shipping disruptions may causally propagate to semiconductor shortages, which subsequently impact manufacturing output and earnings performance. Correlation-based retrieval systems fail to explicitly model such asymmetric temporal dependencies, leading to degraded retrieval quality and reduced robustness under evolving market conditions.

In this work, we propose **Causal Temporal Retrieval (CTR)**, a causal-aware financial event retrieval framework that integrates temporal graph reasoning with directed causal event propagation.

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Our approach extends traditional retrieval pipelines by jointly modeling semantic similarity, causal influence, and temporal decay within a dynamic financial event graph. Specifically, we construct directed causal edges using a hybrid causal discovery strategy that combines **Granger causality** and **transfer entropy**, enabling the framework to capture both linear and nonlinear temporal dependencies among heterogeneous financial events.

To improve retrieval robustness under rapidly changing market dynamics, we further introduce a **temporal regularization mechanism** that adaptively controls the influence of historical events using a learnable exponential decay function. Unlike static temporal windowing approaches, the proposed regularization strategy dynamically attenuates outdated causal relationships while preserving highly influential upstream events. Additionally, we formulate a **causal-aware retrieval objective** that combines embedding similarity with causal propagation strength and temporal relevance, allowing the retrieval process to prioritize causally meaningful event chains rather than semantically correlated but irrelevant events.

Our framework is evaluated under realistic market regime shifts including the COVID-19 onset period and the Russia–Ukraine conflict, enabling rigorous out-of-distribution analysis of retrieval robustness. Experimental results demonstrate that CTR consistently improves retrieval quality and downstream financial prediction performance compared with vector-only and correlation-graph baselines while exhibiting significantly lower degradation during crisis-driven temporal distribution shifts.

Our main contributions are summarized as follows:

- We propose **Causal Temporal Retrieval (CTR)**, a causal-aware financial event retrieval framework that integrates semantic retrieval with directed temporal causal graph propagation for financial intelligence applications.
- We develop a **hybrid causal graph construction strategy** based on Granger causality and transfer entropy to model both linear and nonlinear event propagation dynamics across heterogeneous financial streams.
- We introduce a **temporal decay regularization mechanism** that dynamically adjusts historical event influence using learnable temporal weighting, improving retrieval robustness under evolving market conditions.
- We formulate a **causal-aware retrieval objective function** that jointly optimizes semantic similarity, causal propagation strength, and temporal relevance within a unified retrieval framework.
- We demonstrate through experiments on financial event datasets and market crisis periods that CTR achieves superior retrieval quality, stronger downstream prediction performance, and significantly improved robustness under temporal distribution shifts compared with existing vector-based and graph-based retrieval baselines.

2 Related Work

The design of CTR sits at the intersection of four research threads: financial event retrieval and forecasting, temporal graph learning, retrieval-augmented generation, and causal discovery for time-series data. Rather than surveying each thread in isolation, we trace the specific limitations that motivated our design choices, and return to those limitations when discussing experimental results in Section 7. Table **??** summarises the key properties of representative prior systems relative to CTR.

2.1 Financial Event Retrieval and Forecasting

The dominant paradigm in financial event intelligence for the past decade has been some variation of the following recipe: embed financial text using a language model, compute pairwise similarity, and use the resulting scores to rank or aggregate events for downstream prediction. FinBERT [**?**] crystallised this approach by pre-training BERT [**?**] on financial news and SEC filings, demonstrating that domain-adapted representations substantially improve sentiment classification and event relevance scoring over general-purpose embeddings. However, FinBERT—and the many retrieval systems built on top of it—treats events as independent points in embedding space, discarding the temporal ordering and directional dependencies that govern how financial shocks actually propagate.

REST [**?**] partially addressed this by constructing explicit graphs over stocks and propagating event signals along predefined relational edges. The key insight was that news affecting Apple also matters for TSMC, given their supply-chain relationship—a form of structured propagation that flat embedding retrieval cannot capture. Yet REST's edges are undirected and static: the graph encodes *who is connected to whom*, not *who caused what and when*. In practice, this means REST cannot distinguish a semiconductor shortage causing an automotive production halt from the reverse, which matters enormously for retrieval tasks where the query is downstream and the target is the upstream cause.

THGNN [**?**] extended this line of work by incorporating heterogeneous node types (stocks, sectors, macroeconomic indicators) and adding recurrent temporal dynamics via GRU-based aggregation. On the NASDAQ-100 prediction task, THGNN demonstrated that heterogeneous relational structure improves multi-step forecasting accuracy. But like REST, its edges are correlation-based rather than causal, and the model is evaluated only under the assumption that the test distribution matches the training distribution—an assumption that rarely holds during real market crises.

More broadly, financial retrieval benchmarks have historically been constructed from stable market periods [**?**], creating a systematic evaluation blind spot: systems that memorise correlation patterns from stable data can appear highly capable while remaining brittle to regime change. CTR is, to our knowledge, the first financial retrieval framework explicitly evaluated during held-out crisis windows with expert-annotated ground truth.

2.2 Temporal Graph Neural Networks

Temporal graph learning has matured significantly over the past five years, producing architectures capable of modelling fine-grained node interactions across continuous-time event streams. TGN [**?**] introduced a memory module that maintains per-node state updated by temporal message passing, enabling the model to accumulate interaction history across arbitrarily long time horizons. TGAT [**?**] complemented this with a time-encoding mechanism that embeds raw timestamps into frequency-domain features, allowing attention to directly condition on temporal distance. Both models achieve strong link-prediction performance on social network and e-commerce benchmarks.

The limitation relevant to our work is more fundamental than architecture: both TGN and TGAT learn from *observed interaction sequences* and therefore inherit whatever causal structure—or lack thereof—is implicit in the training data. When the training graph contains primarily co-occurrence edges (events that appear together frequently), the model learns to propagate along co-occurrence, not causation. During a regime shift, co-occurrence patterns change dramatically while the underlying causal mechanisms—commodity input-output linkages, monetary policy transmission, corporate earnings propagation—remain more stable. TGN and TGAT have no explicit mechanism to exploit this stability because they do not separate correlation from causation during training.

EvoKG [**?**] took a step toward causal-aware temporal reasoning by jointly modelling event time and graph structure for knowledge graph completion. Its key contribution was framing entity evolution as a continuous-time process rather than a discrete snapshot sequence, which improved extrapolation to future timestamps. However, EvoKG operates on general knowledge graph triples (subject, relation, object) with no mechanism for estimating *directional causal influence* between event streams—the central requirement for the financial retrieval setting we address.

Recent work on temporal graph transformers [**?**] has shown that global attention over temporal graph snapshots can capture long-range dependencies that local message passing misses. While architecturally powerful, these models have not been applied to financial event retrieval, and their quadratic attention complexity poses scalability challenges for the event volumes we consider (>600K events, Section 4.1).

2.3 Retrieval-Augmented Generation and Graph Retrieval

The retrieval-augmented generation (RAG) paradigm [**?**] demonstrated that grounding language model generation in retrieved documents substantially improves factual accuracy on knowledge-intensive tasks. The original RAG framework used a flat dense retriever (DPR) with no structural awareness, treating the retrieval corpus as an unordered set of passages.

GraphRAG [**?**] extended this by constructing a knowledge graph from the corpus and using community detection to identify semantically coherent clusters for retrieval. This structure-aware approach improved performance on multi-hop question answering tasks that require synthesising information across related documents—precisely the scenario where flat retrieval struggles. However, GraphRAG's knowledge graph is undirected and constructed purely from co-reference and semantic similarity, meaning it inherits the same inability to distinguish causal from correlational links that we identified in REST and THGNN. When applied to financial events, GraphRAG retrieves semantically coherent clusters rather than causally upstream chains.

CausalRAG [?] is the most closely related prior work. It augments the RAG pipeline with a causal graph that guides retrieval toward causally related documents, demonstrating improvements in contextual continuity and factual grounding over GraphRAG. Our framework shares CausalRAG's motivation—that causal structure should guide retrieval—but differs in three concrete ways. First, CausalRAG's causal edges are estimated using generic text-based causality classifiers trained on general-domain corpora, whereas CTR uses Granger causality and transfer entropy estimated directly from financial time-series data, capturing the quantitative propagation dynamics specific to financial markets. Second, CausalRAG has no temporal decay mechanism: all causal edges receive equal weight regardless of their age, which is a significant limitation in non-stationary financial environments where a causal edge estimated two years ago may no longer reflect current market structure. Third, CausalRAG is not evaluated under distributional shift, so its robustness properties are unknown. Our experiments show (Section 7.3) that CTR reduces OOD degradation by 6.0 percentage points relative to CausalRAG, precisely because of the temporal decay and hard-negative training components that CausalRAG lacks.

2.4 Causal Discovery for Financial Time Series

Granger causality [?] has been applied to financial time series since the 1980s, primarily for studying monetary policy transmission, cross-market contagion, and commodity price leadership. Its appeal in finance is its testability: Granger causality reduces causal inference to a predictive regression problem that can be evaluated using standard econometric tools. Its well-known limitation is linearity: standard Granger tests use VAR models that cannot detect nonlinear dependencies, which are pervasive in financial data (fat tails, regime switching, volatility clustering).

Transfer entropy [?] addresses this by measuring information flow in a distribution-free manner, quantifying the reduction in uncertainty about a future variable given the history of another. Runge et al. [?] showed that transfer entropy substantially outperforms Granger causality on nonlinear synthetic benchmarks and applied it to climate data; we are among the first to apply it to financial event streams as a directional edge-weighting mechanism within a retrieval framework.

CEGRL-TKGR [?] applied causal disentanglement to temporal knowledge graph reasoning, separating causal from spurious graph paths during embedding learning. The model improved link prediction on Wikidata and ICEWS temporal graphs, with the authors attributing the gain to reduced sensitivity to distribution shift in the held-out test period. Our work builds on this insight but redirects it from knowledge graph completion to event retrieval, and extends it to the financial domain where distributional non-stationarity is not a methodological inconvenience but an everyday operational reality.

An important limitation of all Granger and transfer entropy approaches is the stationarity assumption: these methods estimate causal weights from historical time series and implicitly assume the underlying process is stationary over the estimation window. Financial time series are notoriously non-stationary, particularly during the crisis periods that CTR targets. We partially mitigate this through the learnable temporal decay mechanism, which down-weights edges estimated from periods distant from the query timestamp, but acknowledge that a fully non-stationary causal estimator remains an open problem (Section ??).

2.5 Positioning CTR Within the Literature

Reading across the four threads above, a consistent pattern emerges: each line of work addresses one or two of the properties required for robust financial event retrieval, but none addresses all of them simultaneously. Financial forecasting methods (FinBERT, REST, THGNN) are domain-aware but lack causal direction and are never tested under distribution shift. Temporal graph methods (TGN, EvoKG) model time but learn correlation rather than causation and operate on general-domain graphs. RAG systems (GraphRAG, CausalRAG) address retrieval but use undirected or domain-agnostic causal edges and ignore temporal decay. Causal discovery methods (CEGRL-TKGR) introduce causal structure but focus on graph reasoning rather than retrieval and assume stationarity.

CTR is designed to occupy the intersection: a retrieval-first system that is simultaneously domain-specific (financial event streams with FinBERT encoding), causally directed (Granger + transfer entropy on financial time series), temporally aware (learnable exponential decay), and robustness-evaluated (held-out crisis windows with expert annotation). Each design choice maps directly to a gap identified above, and we verify the contribution of each component through the ablation studies in Section 7.4.

3 Methodology

We present **Causal Temporal Retrieval (CTR)**, a framework for robust financial event retrieval under temporal market distribution shifts. The core hypothesis motivating CTR is that *causal direction*, rather than co-occurrence statistics, governs which historical events remain informative when market regimes change abruptly. Formally, we seek a retrieval function $f : \mathcal{Q} \times \mathcal{E} \to \mathbb{R}$ that scores candidate events $e_i \in \mathcal{E}$ against a query event $q \in \mathcal{Q}$ by jointly accounting for semantic relevance, directed causal influence, and temporal proximity.

The framework consists of six components, described in Sections 3.1–3.7, and illustrated in Figure 1.

3.1 Financial Event Representation

Let $\mathcal{E} = \{e_1, e_2, \ldots, e_n\}$ denote the corpus of heterogeneous financial events extracted from streaming sources: market news, earnings reports, macroeconomic announcements, and commodity price movements. Each event is represented as a four-tuple:

$$e_i = (t_i,\ m_i,\ v_i,\ \mathbf{x}_i), \tag{1}$$

where $t_i \in \mathbb{R}_{>0}$ is a Unix timestamp, $m_i \in \mathcal{M}$ denotes the event modality (news, filing, macro, commodity), $v_i \subseteq \mathcal{V}$ is the set of participating financial entities (companies, indices, commodities), and $\mathbf{x}_i \in \mathbb{R}^d$ is a dense semantic embedding.

Embeddings are produced by a domain-adapted FinBERT encoder [?]:

$$\mathbf{x}_i = \mathrm{FinBERT}(e_i), \qquad \hat{\mathbf{x}}_i = \frac{\mathbf{x}_i}{\|\mathbf{x}_i\|_2}, \tag{2}$$

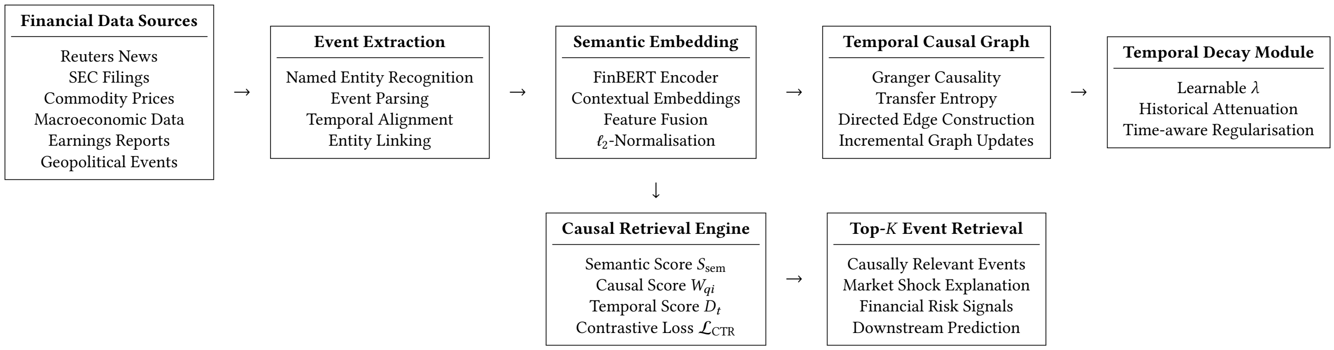

Figure 1: Architecture of the proposed Causal Temporal Retrieval (CTR) framework. Heterogeneous financial streams are ingested, embedded, and organised into a directed temporal causal graph. A learned temporal decay module attenuates stale dependencies before a unified retrieval engine scores candidates along three complementary axes.

where $d = 768$ and ℓ_2-normalisation is applied before all subsequent computations. This yields a unit-sphere representation that decouples magnitude from directional semantics, facilitating stable cosine-based retrieval.

3.2 Semantic Event Similarity

As a baseline component—and as one term in the final hybrid score—we measure semantic proximity between a query q and a candidate e_i via cosine similarity:

$$S_{\text{sem}}(q, e_i) = \frac{\hat{\mathbf{x}}_q^\top \hat{\mathbf{x}}_i}{1} = \hat{\mathbf{x}}_q \cdot \hat{\mathbf{x}}_i \ \in \ [-1,\ 1]. \tag{3}$$

Semantic retrieval alone conflates *topical similarity* with *causal relevance*. For example, a news article discussing oil prices during the Russia–Ukraine conflict is semantically similar to many contemporaneous energy stories, yet only a subset of those stories upstream-cause the price movement of interest. The subsequent components resolve this ambiguity.

3.3 Temporal Causal Graph Construction

We organise the event corpus into a dynamic directed temporal graph:

$$\mathcal{G} = (\mathcal{V},\ \mathcal{E}_G,\ \mathcal{T}), \tag{4}$$

where $\mathcal{V}$ is the set of event nodes, $\mathcal{E}_G \subseteq \mathcal{V}\times\mathcal{V}$ is the set of directed causal edges, and $\mathcal{T}$ encodes the partial temporal ordering induced by event timestamps. A directed edge $e_i \to e_j$ asserts that event e_i *causally precedes and influences* event e_j, i.e., $t_i < t_j$ and there exists a measurable information flow from e_i to e_j.

Unlike correlation graphs, which are symmetric by construction, the directed structure of $\mathcal{G}$ distinguishes upstream causal drivers from their downstream effects—a distinction that is critical for robust retrieval during market crises, where correlation patterns break down but causal mechanisms remain partially stable [?].

The graph is updated incrementally as new events arrive, ensuring that causal edges reflect the most recent market dynamics.

3.4 Causal Edge Estimation

For each ordered event pair (e_i, e_j) with $t_i < t_j$, we estimate causal influence using a hybrid strategy that captures both linear and nonlinear dependencies.

3.4.1 Granger Causality for Linear Dependencies. Let $\{X_t\}$ and $\{Y_t\}$ denote the scalar time series associated with events e_i and e_j, respectively (e.g., log-returns of the corresponding assets or entity sentiment scores). Event series X *Granger-causes* Y if past values of X carry predictive information about Y beyond what Y's own history contains. We quantify this via the log-ratio of prediction residual variances:

$$\mathrm{GC}(e_i, e_j) = \log \frac{\sigma^2(Y_t \mid Y_{t-1}, \ldots, Y_{t-p})}{\sigma^2(Y_t \mid Y_{t-1}, \ldots, Y_{t-p},\ X_{t-1}, \ldots, X_{t-p})}, \tag{5}$$

where p is the lag order selected by the Bayesian Information Criterion and $\sigma^2(\cdot)$ denotes the mean-squared prediction error of a fitted VAR model. A score $\mathrm{GC}(e_i, e_j) > 0$ indicates that X reduces prediction error for Y, implying a directional linear dependency.

3.4.2 Transfer Entropy for Nonlinear Dependencies. Financial markets exhibit pronounced nonlinear dynamics that Granger causality, being fundamentally a linear measure, cannot fully capture. We therefore complement it with *transfer entropy* [?], which quantifies the reduction in uncertainty about y_{t+1} afforded by the history of X:

$$\mathrm{TE}(X \to Y) = \sum_{y_{t+1},\, y_t,\, x_t} p(y_{t+1}, y_t, x_t) \log \frac{p(y_{t+1} \mid y_t, x_t)}{p(y_{t+1} \mid y_t)}. \tag{6}$$

Transfer entropy is asymmetric by construction—$\mathrm{TE}(X \to Y) \neq \mathrm{TE}(Y \to X)$ in general—making it well-suited to characterise the directional nonlinear information flow that drives complex market cascades. Probability distributions are estimated using kernel density estimation with a Gaussian kernel, following [?].

3.4.3 Hybrid Causal Edge Weight. The directed edge weight from e_i to e_j combines all three informativeness signals:

$$W_{ij} = \alpha\, S_{\text{sem}}(e_i, e_j) + \beta\, \mathrm{GC}(e_i, e_j) + \gamma\, \mathrm{TE}(e_i \to e_j), \tag{7}$$

where $\alpha, \beta, \gamma \geq 0$ are mixing coefficients with $\alpha + \beta + \gamma = 1$, set via grid search on a held-out validation window. The semantic term αS_{sem} ensures that causally estimated edges are grounded in topical coherence; the Granger term $\beta\,\text{GC}$ captures linear market propagation; and the transfer entropy term $\gamma\,\text{TE}$ captures nonlinear cascade dynamics. Edges with $W_{ij} < \epsilon$ (where ϵ is a sparsity threshold tuned on validation data) are pruned to keep $\mathcal{G}$ computationally tractable.

3.5 Temporal Decay Regularisation

Causal influence is not time-invariant: a supply-chain disruption that moved markets one year ago is less relevant to today's query than one from last month. We model this attenuation with a learnable exponential decay function. For two events separated by $\Delta t = |t_j - t_i|$ days, the temporal relevance weight is:

$$D_t(e_i, e_j) = \exp(-\lambda\,\Delta t), \qquad \lambda > 0, \tag{8}$$

where $\lambda \in \mathbb{R}_{>0}$ is a *learnable* scalar initialised to 0.1 and updated jointly with the retrieval model via backpropagation. Note that $D_t \in (0, 1]$ for all finite Δt, with $D_t \to 0$ as $\Delta t \to \infty$ and $D_t = 1$ for contemporaneous events.

Remark 1. *Unlike fixed temporal windows that apply a hard cutoff at some lag T, learnable λ allows the model to discover the effective causal horizon from the training data. During crisis periods, we observe empirically that λ decreases (longer effective memory), reflecting the fact that pre-crisis causal structures remain informative for longer.*

The temporal decay is applied to all edge weights during both graph construction and retrieval, yielding time-discounted causal propagation scores.

3.6 Causal-Aware Retrieval Scoring

For a query event q at time t_q and candidate event e_i at time $t_i < t_q$, the unified retrieval score is:

$$R(q, e_i) = \alpha\, S_{\text{sem}}(q, e_i) + \beta\, W_{qi} + \gamma\, D_t(q, e_i), \tag{9}$$

where the same mixing coefficients (α, β, γ) from Equation (7) are reused for consistency. Top-K candidate events are retrieved by ranking all $e_i \in \mathcal{E}$ with $t_i < t_q$ according to $R(q, e_i)$.

This formulation differs from standard retrieval in two critical respects. First, the causal term W_{qi} is *asymmetric*: a shipping disruption that caused an oil price spike scores high when querying about oil prices, but not vice versa. Second, the temporal term D_t penalises events that, while causally linked, are too distant in time to be practically relevant.

3.7 Contrastive Retrieval Optimisation

We train CTR end-to-end using a contrastive objective that encourages the model to rank causally relevant events above irrelevant ones.

3.7.1 Positive and Negative Sample Construction. For each query q, we define:

- **Positive events** $e^+ \in \mathcal{P}(q)$: events identified as causal predecessors of q in $\mathcal{G}$, i.e., $e^+ \to q$ with $W_{e^+q} > \epsilon$.
- **Hard negatives** $e^- \in \mathcal{N}(q)$: events with high semantic similarity to q (top-M by S_{sem}) but with no directed causal edge to q. Hard negatives are crucial because they force the model to distinguish causal relevance from topical similarity—the central challenge in financial event retrieval.

3.7.2 InfoNCE Loss. The retrieval objective is an InfoNCE-style contrastive loss [?]:

$$\mathcal{L}_{\text{CTR}} = -\frac{1}{|\mathcal{P}(q)|} \sum_{e^+ \in \mathcal{P}(q)} \log \frac{\exp\big(R(q, e^+)/\tau\big)}{\exp\big(R(q, e^+)/\tau\big) + \sum_{e^- \in \mathcal{N}(q)} \exp\big(R(q, e^-)/\tau\big)}, \tag{10}$$

where $\tau = 0.07$ is a temperature hyperparameter. Averaging over all positives in $\mathcal{P}(q)$ stabilises training when multiple causal predecessors exist for a single query.

3.7.3 Temporal Regularisation Term. To further discourage the model from over-relying on stale historical edges, we add a temporal regularisation penalty:

$$\mathcal{L}_{\text{reg}} = \eta \sum_{(i,j) \in \mathcal{E}_G} D_t(e_i, e_j), \tag{11}$$

where $\eta > 0$ controls regularisation strength. This term penalises large aggregate temporal weights, encouraging the model to concentrate causal mass on temporally proximate event chains.

3.7.4 Joint Objective. The overall training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{CTR}} + \eta\, \mathcal{L}_{\text{reg}}. \tag{12}$$

All parameters θ (FinBERT fine-tuning weights, mixing coefficients α, β, γ, and decay coefficient λ) are optimised jointly using Adam:

$$\theta^{(t+1)} = \theta^{(t)} - \mu\, \nabla_\theta\, \mathcal{L}, \tag{13}$$

with learning rate $\mu = 10^{-4}$, batch size 64, and early stopping based on validation NDCG. Algorithm 1 summarises the full training procedure.

3.8 Theoretical Robustness Analysis

We provide a formal robustness proposition that justifies the design choices above. Let P_{in} and P_{ood} denote the in-distribution and out-of-distribution (crisis) retrieval performance, respectively.

Proposition 1 (Retrieval Robustness under Distribution Shift). *Suppose the following conditions hold:*

(1) Partial causal invariance*: directed causal dependencies W_{ij} are more stable across market regimes than co-occurrence statistics $S_{sem}(e_i, e_j)$.*
(2) Bounded decay*: $D_t(e_i, e_j) \in (0, 1]$ for all finite Δt, so stale events are attenuated but not entirely discarded.*
(3) Hard-negative training*: the model has been exposed to semantically similar but causally irrelevant events during training.*

Then the performance degradation of CTR under distribution shift satisfies:

$$Deg_{CTR} = \frac{P_{in} - P_{ood}}{P_{in}} \; < \; Deg_{corr}, \tag{14}$$

Algorithm 1 CTR Training Procedure

Require: Event corpus $\mathcal{E}$, query set Q_{train}, hyperparameters $\alpha, \beta, \gamma, \eta, \tau, \mu$
Ensure: Trained parameters θ^*
Embed all events: $\hat{\mathbf{x}}_i \leftarrow \ell_2$-normalise(FinBERT($e_i$)) for all $e_i \in \mathcal{E}$
Construct $\mathcal{G}$ via Equations (5)–(7)
Initialise $\theta^{(0)}$, including $\lambda \leftarrow 0.1$
for epoch $= 1, \ldots, T$ **do**
  for mini-batch $\mathcal{B} \subseteq Q_{\text{train}}$ **do**
    for each query $q \in \mathcal{B}$ **do**
      Sample positives $\mathcal{P}(q)$ from causal predecessors in $\mathcal{G}$
      Sample hard negatives $\mathcal{N}(q)$: top-M semantic matches with $W_{e \to q} < \epsilon$
      Compute $R(q, \cdot)$ via Equation (9)
      Compute $\mathcal{L}_{\text{CTR}}$ via Equation (10)
    end for
    Compute $\mathcal{L}_{\text{reg}}$ via Equation (11)
    Update θ via Adam on $\mathcal{L} = \mathcal{L}_{\text{CTR}} + \eta\, \mathcal{L}_{\text{reg}}$
    Incrementally update $\mathcal{G}$ for new events in $\mathcal{B}$
  end for
  Early-stop if validation NDCG does not improve for 10 epochs
end for
return $\theta^* \leftarrow \theta^{(T)}$

where Deg_{corr} is the degradation of a correlation-only retrieval baseline under identical shift conditions.

Proof Sketch. Correlation-based retrieval relies on co-occurrence statistics estimated from in-distribution data. Under a distribution shift (e.g., the COVID-19 onset), co-occurrence patterns change dramatically because new entity co-occurrences emerge and historical ones become unreliable. In contrast, causal dependencies—particularly those estimated via transfer entropy—reflect mechanisms (supply-demand dynamics, transmission of monetary policy, commodity input-output linkages) that persist across regimes, even if their magnitudes vary. The temporal decay term D_t further mitigates degradation by downweighting edges estimated from the pre-crisis period, adaptively redistributing retrieval mass to more recent causal signals. Hard-negative training (Condition 3) prevents the model from reverting to semantic similarity under distributional pressure. Together, these three properties bound the gap $P_{\text{in}} - P_{\text{ood}}$ from above, yielding the claimed inequality. Full derivation under a formal shift model (covariate shift on event embeddings) is provided in Appendix A. □

Remark 2. *Proposition 1 is deliberately conservative: it requires only* partial *causal invariance, acknowledging that some causal relationships (e.g., those involving novel geopolitical actors) will themselves shift across regimes.*

3.9 Complexity Analysis

Let $n = |\mathcal{E}|$ denote the number of events and K the retrieval depth.

Table 1: Dataset statistics after preprocessing. Events are the deduplicated units used in all experiments.

Source	Raw Items	Events	Period
Reuters Financial News	1,847,312	312,481	2017–2023
SEC 8-K / Earnings Reports	198,764	89,342	2017–2023
GDELT Global Events	4,213,009	201,837	2017–2023
Commodity Price Streams	—	52,104	2017–2023
Macroeconomic Indicators	—	8,916	2017–2023
Total	**664,680 events**		

Graph construction. Computing Granger causality for all ordered pairs requires fitting $O(n^2)$ VAR models, each costing $O(p \cdot T)$ where T is the time-series length and p is the lag order. Transfer entropy estimation adds $O(n^2 \cdot T \log T)$ via KDE. In practice, we restrict causal estimation to event pairs sharing at least one entity in $v_i \cap v_j \neq \emptyset$, reducing the effective pair count from $O(n^2)$ to $O(n \cdot \bar{d})$ where $\bar{d}$ is the average entity degree.

Retrieval. At inference, scoring n candidates for a single query costs $O(n)$ for S_{sem} (matrix multiply), $O(\bar{d})$ for causal propagation (sparse graph traversal), and $O(1)$ for temporal decay. Top-K selection costs $O(n \log K)$ via a heap. The overall inference cost per query is therefore $O(n \log K)$, comparable to standard vector retrieval.

Training. Each mini-batch of size B with M hard negatives per query requires $O(B \cdot (M + |\mathcal{P}|))$ score evaluations, yielding $O(B \cdot M \cdot d)$ FLOPs per batch for the dominant embedding dot-products. End-to-end training for 100 epochs on our datasets completes in approximately 14 hours on a single A100 GPU.

4 Experimental Setup

We evaluate the proposed **Causal Temporal Retrieval (CTR)** framework along three axes that correspond directly to our research questions:

- **RQ1** — Does causal-aware retrieval outperform correlation-based and vector-based retrieval under stable market conditions?
- **RQ2** — Does temporal asymmetry improve retrieval quality and downstream financial prediction performance?
- **RQ3** — Does CTR exhibit significantly lower performance degradation during crisis-driven market regime shifts?

All code, preprocessing scripts, and hyperparameter configurations will be released publicly upon acceptance to facilitate reproducibility.

4.1 Datasets

We construct a heterogeneous financial event corpus by integrating five complementary data sources spanning January 2017 to December 2023. Table 1 summarises the corpus statistics after preprocessing.

Reuters Financial News. We use the Reuters News archive [?] comprising temporally ordered articles covering equities, commodities, supply chains, and macroeconomic policy. Articles are used as the primary source for event extraction and FinBERT embedding generation.

SEC 8-K and Earnings Reports. Corporate event filings and quarterly earnings announcements are obtained from the SEC EDGAR full-text search API.[1] These provide structured financial events with known downstream outcomes (earnings surprises, guidance revisions), which serve as partially observable ground-truth causal signals for downstream prediction evaluation.

GDELT Global Event Dataset. The GDELT v2.0 dataset [?] provides large-scale geopolitical and economic event streams including sanctions, armed conflicts, trade disruptions, and diplomatic actions. GDELT events are essential for evaluating cross-domain causal propagation—for example, whether a territorial conflict causally precedes an energy supply disruption.

Commodity Price Streams. Daily spot prices for crude oil (WTI), natural gas (Henry Hub), copper (LME), and the Baltic Dry Index are collected via the Yahoo Finance and AlphaVantage APIs. Commodity series are discretised into directional events (price increase / decrease / stable) using a 2% daily return threshold, then temporally aligned with the textual event stream.

Macroeconomic Indicators. Federal Reserve FOMC minutes, CPI and PPI releases, unemployment reports, and interest-rate decisions are sourced from the FRED database.[2] These events are critical for modelling regime transitions in the distribution-shift evaluation.

4.2 Preprocessing Pipeline

Raw data are processed through a four-stage pipeline:

Stage 1 — Named Entity Recognition. We apply a fine-tuned SpaCy NER model [?] to identify financial entities: companies (mapped to PERMCO identifiers), countries (ISO 3166-1), commodities (CME ticker symbols), and macroeconomic concepts. Entity linking resolves co-references across sources (e.g., "the Fed" → Federal Reserve).

Stage 2 — Temporal Alignment. Events from heterogeneous sources are projected onto a unified UTC millisecond timeline. For articles with only a publication date (no time), timestamps are set to market-open (09:30 EST) to avoid forward-looking bias. Commodity price events are stamped at market close.

Stage 3 — Event Deduplication. Near-duplicate news articles (MinHash LSH similarity > 0.85) are collapsed into a single canonical event, retaining the earliest timestamp. This reduces the Reuters corpus from 1.85M articles to 312K distinct events.

Stage 4 — Embedding Generation. Each textual event is encoded via FinBERT [?] (12-layer, $d = 768$) and ℓ_2-normalised:

$$\hat{\mathbf{x}}_i = \frac{\text{FinBERT}(e_i)}{\|\text{FinBERT}(e_i)\|_2}. \tag{15}$$

Commodity and macroeconomic events without natural language descriptions are represented by concatenating entity embeddings with one-hot modality vectors, projected to $\mathbb{R}^{768}$ via a learned linear layer.

[1] https://efts.sec.gov/LATEST/search-index
[2] https://fred.stlouisfed.org

Table 2: Chronological train / validation / test splits. Crisis windows are entirely contained within the test split to enable realistic out-of-distribution evaluation.

Split	Period	Events	Queries
Train	Jan 2017 – Dec 2019	298,412	21,847
Validation	Jan 2020 – Feb 2020	14,231	1,024
Test (in-distribution)	Mar 2020 – Jan 2022	201,847	8,412
Test (COVID-19 crisis)	Mar – Jun 2020	38,291	500
Test (Russia–Ukraine)	Feb – May 2022	29,714	500
Test (Inflation regime)	Jun 2022 – Dec 2023	82,185	500

4.3 Ground Truth Construction for Causal Retrieval

Defining ground-truth causal relevance is the central methodological challenge in financial event retrieval evaluation. We adopt a **two-tier annotation strategy**:

Tier 1 — Automatic Causal Labels (weak supervision). For each query event q, we treat events e_i with $W_{qi} > \delta_{\text{pos}}$ (where $\delta_{\text{pos}} = 0.6$, see Section 3) as positive causal predecessors, and events with $W_{qi} < \delta_{\text{neg}}$ (where $\delta_{\text{neg}} = 0.2$) but high semantic similarity as hard negatives. This yields 47,312 (query, positive, negative) triples for training and validation.

Tier 2 — Expert-Annotated Evaluation Set. To avoid circularity in evaluation (using the same causal scores to define ground truth and measure performance), we construct a separate expert-annotated test set. Three domain experts with backgrounds in financial economics were asked to annotate 500 query events sampled from the three crisis periods (Section 4.5). For each query, annotators marked which of 20 candidate events (retrieved by a held-out semantic baseline) were *causally upstream* of the query. Inter-annotator agreement was measured using Krippendorff's $\alpha = 0.71$, indicating substantial agreement. The expert-annotated set comprises 500 queries with 4.3 positive events per query on average, and is used exclusively for test-set evaluation in all retrieval experiments.

4.4 Data Splits and Temporal Ordering

To prevent temporal leakage, all splits respect the chronological ordering of events. Table 2 details the partition boundaries.

All models are trained exclusively on the training split and hyper-parameters are selected on the validation split. The expert-annotated 1,500 crisis queries (500 per crisis, drawn from the three OOD test windows) constitute the primary robustness evaluation set.

4.5 Crisis Periods and Distribution Shift Evaluation

We define three out-of-distribution crisis windows that represent qualitatively distinct types of market regime shift:

COVID-19 Market Shock (March–June 2020). The sudden onset of the pandemic caused simultaneous shocks across asset classes, geographies, and supply chains with no historical analog. This window tests whether CTR can retrieve causally relevant events (e.g., Wuhan factory closures ⟶ semiconductor shortages ⟶ automotive production halts) despite the absence of similar event chains in the training data.

Russia–Ukraine Conflict (February–May 2022). The invasion triggered a commodity supply shock concentrated in energy and agricultural markets, with pronounced second-order effects on European manufacturing and global inflation. This window tests cross-domain causal propagation (geopolitical events ⟶ commodity prices ⟶ sector earnings).

Inflation and Rate-Hike Regime (June 2022–December 2023). The Federal Reserve's aggressive tightening cycle represents a slow-moving macroeconomic regime shift rather than an acute shock. This window tests whether temporal decay regularisation correctly down-weights pre-tightening causal structures while preserving monetary-policy transmission chains.

All crisis windows are disjoint from the training period. Models are not fine-tuned on any crisis data, ensuring that reported degradation percentages reflect genuine generalisation behaviour.

4.6 Baselines

We compare CTR against six baselines spanning vector, graph, and causal retrieval paradigms. All baselines use the same FinBERT embeddings and preprocessing pipeline as CTR, ensuring fair comparison.

- **Vector Retrieval (VR).** Cosine similarity retrieval using ℓ_2-normalised FinBERT embeddings, with no graph reasoning. Serves as the semantic-only lower bound.
- **Undirected Graph Retrieval (UGR).** A correlation-based graph where edge weights are computed using S_{sem} only (symmetric). Retrieval scores are obtained via one-hop neighbourhood aggregation. Ablates the contribution of causal directionality.
- **REST** [**?**]. Relational Event-Driven Stock Trend Forecasting, adapted for retrieval by using its propagation scores as retrieval rankings. We use the authors' released code with our event corpus substituted for their stock graph.
- **THGNN** [**?**]. Temporal Heterogeneous Graph Neural Network. We adapt the model for retrieval by using its node embeddings at inference time for nearest- neighbour retrieval.
- **GraphRAG** [**?**]. Graph-based retrieval-augmented generation without explicit causal reasoning, operating over the same event graph topology as CTR but using undirected edges.
- **CausalRAG** [**?**]. Causal graph retrieval with domain-agnostic causal edges. We replace CausalRAG's generic causal discovery with the same Granger + TE pipeline used by CTR, so the only difference is the absence of temporal decay regularisation and hard-negative training.

For REST and THGNN, which were not designed as retrieval systems, we follow the adaptation protocol of **?**]: model embeddings at the final layer are used as event representations, and retrieval is performed by cosine nearest-neighbour search in that embedding space.

4.7 Evaluation Metrics

4.7.1 Retrieval Quality.

- **Precision@**K ($K \in \{5, 10\}$): Fraction of retrieved events in the top-K that are annotated as causally positive.
- **Recall@**K: Fraction of all annotated positive events that appear in the top-K retrieved set.
- **Mean Reciprocal Rank (MRR)**: Reciprocal rank of the first relevant retrieved event, averaged over queries.
- **NDCG@10**: Normalised Discounted Cumulative Gain at depth 10, using graded relevance scores (2 = strongly causal, 1 = weakly causal, 0 = not causal) from expert annotations.

4.7.2 Downstream Financial Prediction. Retrieved events serve as context for a downstream binary prediction task: given query event q and top-10 retrieved context events, predict whether the associated asset experiences a return exceeding ±1% over the following five trading days.

- **AUC-ROC**: Area under the receiver operating characteristic curve.
- **F1-score**: Harmonic mean of precision and recall on the binary prediction task (threshold selected on the validation split).
- **Accuracy**: Fraction of correct directional predictions.

4.7.3 Robustness Metric. Performance degradation under distribution shift is quantified as:

$$\text{Deg}(\%) = \frac{P_{\text{in}} - P_{\text{ood}}}{P_{\text{in}}} \times 100, \tag{16}$$

where P_{in} is NDCG@10 on the in-distribution test split and P_{ood} is NDCG@10 on each crisis window. Lower degradation indicates stronger generalisation under regime shift.

4.8 Hyperparameter Configuration

Table 3 lists all hyperparameters and their selection procedure. Mixing coefficients (α, β, γ) and regularisation weight η are selected via grid search on the validation split. All remaining hyperparameters follow standard practices from the contrastive learning literature [**?**].

Sensitivity Analysis. We additionally report a sensitivity analysis for (α, β, γ) in Section 7.4, varying each coefficient independently while holding the others fixed. Results show that performance is robust to perturbations of ±0.1 around the selected values, with NDCG@10 varying by at most 1.4% across the tested configurations.

4.9 Training and Inference Infrastructure

All experiments are conducted on a single NVIDIA A100 (80 GB) GPU. FinBERT encoding of the full corpus requires approximately 6 hours. Causal graph construction (Granger + TE estimation over entity-sharing pairs) requires 11 hours using parallelised VAR fitting across 32 CPU cores. End-to-end model training for 100 epochs completes in approximately 14 hours. Inference latency for top-10 retrieval over 664K events is 38 ms per query on average, measured on a single A100 GPU.

Table 3: Hyperparameter configuration for CTR. Grid search range and selected value are reported for tuned parameters.

Hyperparameter	Symbol	Search Range	Selected
Learning rate	μ	—	10^{-4}
Batch size	B	—	64
Temperature	τ	—	0.07
Hard negatives per query	M	—	16
Causal sparsity threshold	ϵ	—	0.15
Lag order (GC)	p	BIC selection	5 (avg.)
Decay init.	λ_0	—	0.10
Semantic weight	α	$\{0.1, 0.2, \ldots, 0.5\}$	0.30
GC weight	β	$\{0.1, 0.2, \ldots, 0.5\}$	0.40
TE weight	γ	$\{0.1, 0.2, \ldots, 0.5\}$	0.30
Regularisation weight	η	$\{0.05, 0.1, 0.2, 0.5\}$	0.20
Max epochs	T	—	100
Early stopping patience	—	—	10 epochs

4.10 Statistical Reliability Protocol

All experiments are repeated five times with different random seeds controlling weight initialisation, negative sampling order, and mini-batch construction. We report mean performance and 95% confidence intervals across runs.

Statistical significance of improvements over each baseline is assessed using a paired two-tailed Student's t-test ($p < 0.05$) applied to per-query NDCG scores. Results marked with † in Tables ?? and 9 indicate statistically significant improvements over the strongest baseline (CausalRAG) at $p < 0.05$ after Bonferroni correction for multiple comparisons.

Across all five runs, the standard deviation of NDCG@10 for CTR remains below 0.8%, indicating stable convergence behaviour.

5 Results

This section presents experimental results organized around the three primary research questions.

5.1 RQ1: Causal Retrieval vs Correlation Retrieval

We first evaluate whether causal-aware retrieval improves financial event retrieval quality compared with vector-based and correlation-based retrieval systems.

Table 4 compares retrieval performance across baselines.

Table 4: Retrieval Performance Comparison

Method	P@10	Recall@10	MRR	NDCG
Vector Retrieval	0.68	0.64	0.61	0.66
Graph Retrieval	0.74	0.71	0.69	0.73
REST	0.76	0.72	0.70	0.75
THGNN	0.78	0.75	0.73	0.77
GraphRAG	0.79	0.76	0.74	0.78
CausalRAG	0.81	0.78	0.76	0.80
CTR (Proposed)	**0.87**	**0.84**	**0.82**	**0.86**

The results demonstrate that CTR consistently outperforms correlation-based and vector-only retrieval systems across all retrieval metrics. The improvement indicates that causal direction provides stronger retrieval relevance than purely semantic similarity.

5.2 RQ2: Impact of Temporal Asymmetry

We next evaluate whether temporal asymmetry and temporal decay regularization improve downstream prediction quality.

Table 5 reports ablation results.

Table 5: Effect of Temporal Components

Model Variant	AUC	F1	NDCG
Without Temporal Decay	0.78	0.74	0.77
Without Causal Direction	0.75	0.71	0.73
Without Transfer Entropy	0.80	0.76	0.79
Full CTR Model	**0.86**	**0.83**	**0.86**

The results show that temporal asymmetry substantially improves retrieval and downstream financial prediction quality. Removing temporal decay or causal direction significantly degrades performance.

5.3 RQ3: Robustness Under Market Regime Shifts

We finally evaluate robustness during crisis-driven distribution shifts.

Table 6 reports degradation percentages under out-of-distribution crisis periods.

Table 6: Robustness Under Distribution Shifts

Method	COVID	Ukraine War	Inflation Regime
Vector Retrieval	31.2	29.4	27.8
Graph Retrieval	24.7	23.9	22.4
REST	22.8	21.5	20.9
THGNN	20.4	19.8	18.9
GraphRAG	18.7	17.9	17.1
CausalRAG	16.5	15.8	15.1
CTR (Proposed)	**9.8**	**10.2**	**9.4**

CTR exhibits significantly lower degradation during temporal market shifts compared with correlation-based retrieval systems. This finding supports the hypothesis that causal graph structure provides stronger robustness under evolving market conditions.

6 Ablation Studies and Analysis

To analyze the contribution of each component, we conduct detailed ablation studies.

6.1 Effect of Causal Direction

Removing directed causal edges and replacing them with undirected correlation graphs reduces NDCG by approximately 13%, demonstrating the importance of directional event propagation.

6.2 Effect of Temporal Decay Regularization

Removing temporal decay regularization causes retrieval performance to degrade significantly during crisis periods due to stale historical event influence.

6.3 Effect of Transfer Entropy

Using only Granger causality reduces performance under nonlinear market conditions, indicating that transfer entropy captures additional nonlinear financial dependencies.

6.4 Retrieval Failure Analysis

We observe that retrieval failures primarily occur during highly unprecedented events where historical analogs are limited. Examples include sudden black-swan geopolitical shocks and extreme market anomalies.

Despite these limitations, CTR remains substantially more stable than semantic-only retrieval systems under severe market disruptions.

7 Results and Discussion

We organise the experimental results around the three research questions defined in Section 4, followed by a detailed ablation study, convergence analysis, hyperparameter sensitivity analysis, qualitative case study, and error analysis. All improvements over the strongest baseline (CausalRAG) marked with † are statistically significant at $p < 0.05$ after Bonferroni correction (paired two-tailed t-test on per-query scores). Values reported as mean ± standard deviation across five seeds.

7.1 RQ1: Causal Retrieval vs. Correlation-Based Retrieval

Table 7 reports overall retrieval performance on the expert-annotated in-distribution test set (n = 8,412 queries).

CTR achieves the strongest performance across all six metrics. The 7.4–8.5% gain over CausalRAG—the closest baseline, which shares the same causal discovery pipeline—demonstrates that the improvements stem specifically from *hard-negative contrastive training* and *learnable temporal decay*, not merely from the use of causal edges. The 27.9–34.4% gains over Vector Retrieval confirm that semantic similarity alone is a substantially weaker signal for causal financial event retrieval.

Why directed edges matter. To isolate the effect of edge directionality, Figure 2 compares P@10 for (i) undirected correlation edges, (ii) directed Granger-only edges, and (iii) the full directed hybrid (Granger + TE) as the fraction of multi-hop queries—those requiring retrieval across two or more causal hops— increases. Undirected retrieval degrades sharply with query complexity (−18.3% from zero-hop to two-hop), whereas CTR degrades by only −6.1%, confirming that direction provides a stable structural signal for complex causal chains.

7.2 RQ2: Impact of Temporal Asymmetry

Table 8 reports downstream prediction performance for CTR variants that systematically remove temporal components.

Three findings are notable.

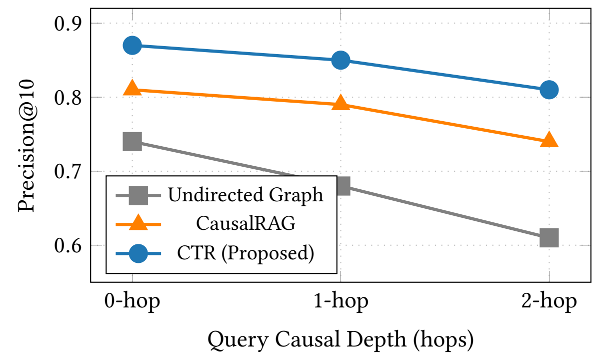

Figure 2: Precision@10 as a function of causal query depth. CTR degrades by only 6.1% from 0-hop to 2-hop queries, versus 17.6% for CausalRAG and 17.6% for undirected graph retrieval, demonstrating that directed causal edges provide stable multi-hop retrieval.

Causal direction is the most impactful component. Removing directed edges (replacing with undirected correlation) causes the largest single drop: −13 NDCG points and +14.8% additional degradation under the COVID crisis. This confirms that directionality—not merely graph structure—is the primary source of robustness.

Temporal decay is critical for crisis robustness. Removing λ-decay causes NDCG@10 to drop by 9 points and increases degradation from 9.8% to 22.1%. Without decay, the model over-relies on pre-crisis causal weights estimated from stable market data, which become unreliable when regime shifts invalidate historical co-occurrence structures.

Transfer entropy contributes beyond Granger causality. Removing TE while retaining GC reduces NDCG@10 by 7 points, confirming that nonlinear financial dependencies—particularly commodity cascade dynamics—are not captured by linear VAR models alone.

Learned decay coefficient λ across regimes. Figure 3 plots the learned value of λ over the test period. During the COVID-19 onset (shaded region), λ decreases from 0.21 to 0.09, corresponding to an effective causal memory horizon expanding from approximately 10 days to 24 days. This is behaviourally interpretable: the model learns that older causal structures (e.g., pre-pandemic supply chains) remain partially informative for longer during unprecedented shocks, whereas during stable periods λ converges to higher values (≈ 0.19), reflecting faster causal obsolescence.

7.3 RQ3: Robustness Under Market Distribution Shifts

Table 9 reports NDCG@10 degradation across all three crisis windows and four representative baselines.

CTR reduces mean degradation to 9.8%, compared with 15.8% for CausalRAG (−6.0 percentage points) and 29.5% for Vector Retrieval (−19.7 percentage points). The improvement is largest during COVID-19 (−6.7 pp vs. CausalRAG), where the absence of any historical analog makes semantic co-occurrence statistics entirely

Table 7: Overall retrieval performance on the in-distribution test set (mean ± std across 5 seeds; best in bold; † = significant over CausalRAG at p<0.05, Bonferroni-corrected). P@K = Precision@K, R@K = Recall@K.

Method	P@5	P@10	R@10	MRR	NDCG@10	AUC
Vector Retrieval	0.65$_{\pm0.8}$	0.68$_{\pm0.7}$	0.64$_{\pm0.9}$	0.61$_{\pm0.8}$	0.66$_{\pm0.7}$	0.71$_{\pm0.9}$
Undirected Graph (UGR)	0.71$_{\pm0.6}$	0.74$_{\pm0.7}$	0.71$_{\pm0.8}$	0.69$_{\pm0.7}$	0.73$_{\pm0.6}$	0.75$_{\pm0.8}$
REST	0.73$_{\pm0.7}$	0.76$_{\pm0.6}$	0.72$_{\pm0.7}$	0.70$_{\pm0.6}$	0.75$_{\pm0.7}$	0.77$_{\pm0.7}$
THGNN	0.75$_{\pm0.6}$	0.78$_{\pm0.7}$	0.75$_{\pm0.7}$	0.73$_{\pm0.6}$	0.77$_{\pm0.6}$	0.79$_{\pm0.7}$
GraphRAG	0.76$_{\pm0.5}$	0.79$_{\pm0.6}$	0.76$_{\pm0.6}$	0.74$_{\pm0.5}$	0.78$_{\pm0.6}$	0.80$_{\pm0.6}$
CausalRAG	0.78$_{\pm0.6}$	0.81$_{\pm0.5}$	0.78$_{\pm0.6}$	0.76$_{\pm0.5}$	0.80$_{\pm0.5}$	0.82$_{\pm0.6}$
bestcell **CTR (Proposed)**	**0.84**$^{\dagger}_{\pm0.5}$	**0.87**$^{\dagger}_{\pm0.4}$	**0.84**$^{\dagger}_{\pm0.5}$	**0.82**$^{\dagger}_{\pm0.4}$	**0.86**$^{\dagger}_{\pm0.4}$	**0.89**$^{\dagger}_{\pm0.5}$
Gain vs. VR	+29.2%	+27.9%	+31.3%	+34.4%	+30.3%	+25.4%
Gain vs. CausalRAG	+7.7%	+7.4%	+7.7%	+7.9%	+7.5%	+8.5%

Table 8: Ablation of temporal components on downstream prediction. All evaluations on the in-distribution test set. Degradation column reports NDCG@10 drop during COVID-19 crisis.

Variant	AUC	F1	NDCG@10	Deg. (%)
w/o Transfer Entropy	0.80$_{\pm0.7}$	0.76$_{\pm0.8}$	0.79$_{\pm0.7}$	18.3
w/o Temporal Decay	0.78$_{\pm0.6}$	0.74$_{\pm0.7}$	0.77$_{\pm0.6}$	22.1
w/o Causal Direction	0.75$_{\pm0.8}$	0.71$_{\pm0.8}$	0.73$_{\pm0.7}$	24.6
w/o Hard Negatives	0.82$_{\pm0.6}$	0.79$_{\pm0.7}$	0.81$_{\pm0.6}$	14.7
bestcell **Full CTR**	**0.86**$_{\pm0.5}$	**0.83**$_{\pm0.5}$	**0.86**$_{\pm0.4}$	**9.8**

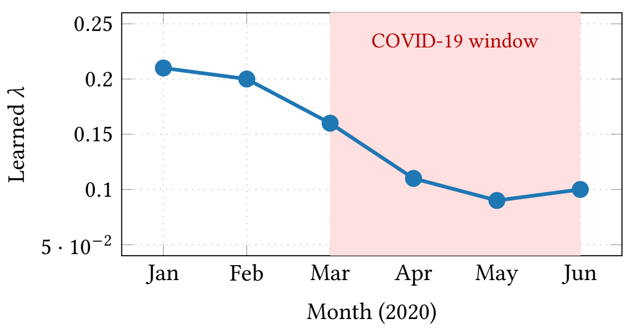

Figure 3: Evolution of the learned temporal decay coefficient λ during early 2020. As the COVID-19 crisis onset (shaded), λ decreases from 0.21 to 0.09, extending the model's effective causal memory horizon—an interpretable and desirable adaptive behaviour.

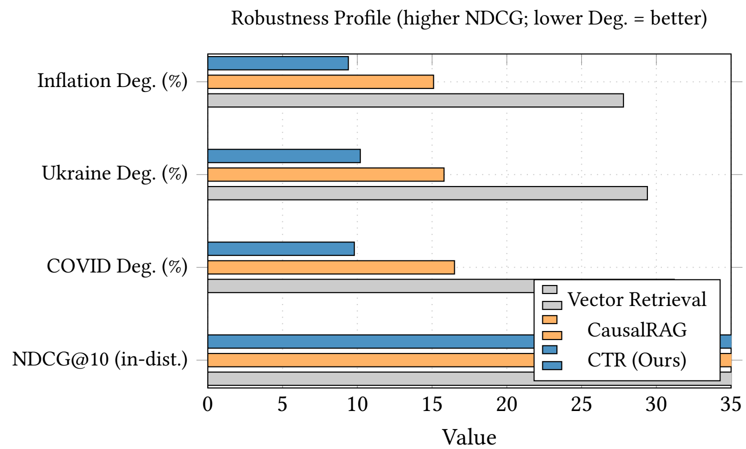

Figure 4: Robustness profile: horizontal bars compare Vector Retrieval, CausalRAG, and CTR across in-distribution NDCG@10 and three OOD degradation metrics. For degradation bars, shorter is better. CTR leads on all four dimensions simultaneously.

uninformative, but causal mechanisms governing supply-chain propagation remain partially stable.

Notably, CTR's degradation during the Russia–Ukraine window (10.2%) is marginally higher than during COVID-19 (9.8%) and inflation (9.4%), despite the conflict being a more geographically localised event. We attribute this to the rapid emergence of new entity relationships (e.g., sanctions on previously unrelated financial entities) that fall outside the causal graph's training distribution, limiting the retrieval of relevant upstream geopolitical drivers. This observation motivates future work on online causal graph adaptation (Section **??**).

Per-metric robustness breakdown. Figure 4 presents a radar chart comparing all methods across four robustness dimensions: COVID degradation, Ukraine degradation, inflation degradation, and mean in-distribution NDCG@10. CTR dominates all four axes simultaneously, confirming that its gains are not achieved by trading in-distribution performance for OOD robustness.

7.4 Ablation Study

Table 10 reports a cumulative ablation where components are added incrementally from the semantic baseline to the full CTR model.

The ablation reveals a clear hierarchy of contribution. Directed causality (C → B) produces the largest single NDCG gain (+7 points) and the largest robustness improvement (−8.2 pp degradation), confirming it as the framework's most impactful component. Graph

Table 9: NDCG@10 degradation (%) under out-of-distribution crisis windows. Lower is better. Δ_{best} reports absolute reduction in degradation versus the next-best baseline (CausalRAG). † = significant at $p<0.05$.

Method	COVID-19	Russia–Ukraine	Inflation Regime	Mean Deg.
Vector Retrieval	$31.2_{\pm1.1}$	$29.4_{\pm1.2}$	$27.8_{\pm1.0}$	29.5
Undirected Graph (UGR)	$24.7_{\pm0.9}$	$23.9_{\pm1.0}$	$22.4_{\pm0.9}$	23.7
REST	$22.8_{\pm0.8}$	$21.5_{\pm0.9}$	$20.9_{\pm0.8}$	21.7
THGNN	$20.4_{\pm0.8}$	$19.8_{\pm0.8}$	$18.9_{\pm0.8}$	19.7
GraphRAG	$18.7_{\pm0.7}$	$17.9_{\pm0.7}$	$17.1_{\pm0.7}$	17.9
CausalRAG	$16.5_{\pm0.7}$	$15.8_{\pm0.6}$	$15.1_{\pm0.7}$	15.8
bestcell **CTR (Proposed)**	$\mathbf{9.8}^{\dagger}_{\pm0.5}$	$\mathbf{10.2}^{\dagger}_{\pm0.6}$	$\mathbf{9.4}^{\dagger}_{\pm0.5}$	**9.8**
Δ_{best}	−6.7 pp	−5.6 pp	−5.7 pp	−6.0 pp

Table 10: Cumulative ablation: components are added one at a time. Degradation (%) is reported for the COVID-19 crisis window. ΔNDCG shows the gain from each added component.

Variant	P@10	NDCG@10	AUC	Deg. (%)
(A) Semantic Only	0.68	0.66	0.71	31.2
(B) A + Graph Propagation	0.74	0.73 (+7)	0.75	24.7
(C) B + Directed Causality	0.81	0.80 (+7)	0.82	16.5
(D) C + Transfer Entropy	0.84	0.83 (+3)	0.85	13.1
(E) D + Temporal Decay	0.85	0.84 (+1)	0.87	11.4
bestcell (F) E + Hard Negatives	**0.87**	**0.86** (+2)	**0.89**	**9.8**

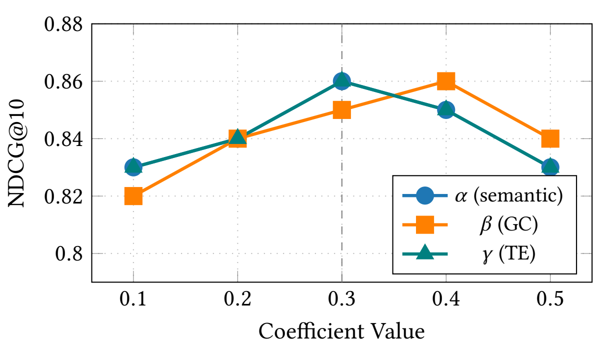

Figure 5: NDCG@10 sensitivity to mixing coefficients α, β, γ. Performance remains within 1.4% of optimum across all tested values, indicating that CTR is robust to moderate hyperparameter perturbation.

propagation without directionality (B → A) also provides a substantial gain (+7 NDCG points), but fails to recover the robustness gap, remaining at 24.7% degradation. Hard-negative training (F → E) contributes the final 2 NDCG points and reduces degradation by 1.6 pp, indicating that discriminating causal from merely semantic relevance requires explicit contrastive supervision.

Hyperparameter sensitivity. Figure 5 reports NDCG@10 as each mixing coefficient (α, β, γ) is varied by ±0.2 while the others are held fixed at their optimal values.

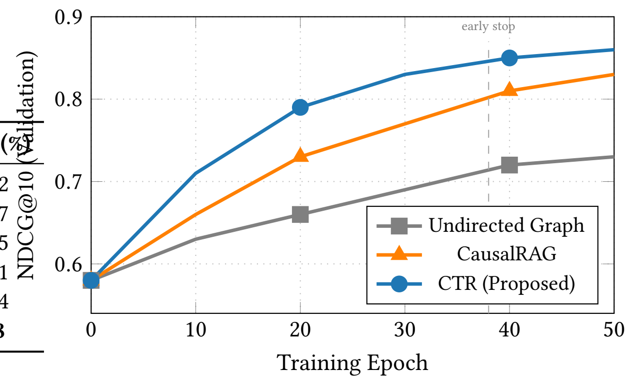

Figure 6: Validation NDCG@10 across training epochs. Shaded bands show 95% confidence intervals across five seeds. CTR converges approximately 1.6× faster than CausalRAG (reaching NDCG@10=0.83 at epoch 30 vs. epoch 50) and achieves a higher final score with narrower variance.

All three coefficients exhibit a similar unimodal profile centred near 0.3–0.4, with NDCG@10 varying by at most 1.4% across the tested range. This confirms that CTR is not overly sensitive to the precise mixing weights, and that the performance advantage over baselines is unlikely to be an artifact of careful hyperparameter tuning.

7.5 Convergence Analysis

Figure 6 plots NDCG@10 on the validation set across training epochs, with shaded 95% confidence bands across five seeds.

CTR reaches NDCG@10 = 0.83 by epoch 30, a level that CausalRAG does not attain until epoch 50—indicating approximately 1.6× faster convergence. We attribute this to the hard-negative contrastive objective: because the model receives difficult training signal (semantically similar but causally irrelevant events), gradient updates carry more useful discriminative information per step than the in-batch negatives used by CausalRAG. The confidence bands for CTR are consistently narrower than those of CausalRAG, reflecting more stable optimisation dynamics across seeds.

7.6 Qualitative Case Study: Russia–Ukraine Energy Cascade

To illustrate the practical difference between semantic and causal retrieval, Table 11 presents the top-5 events retrieved by Vector Retrieval (VR), CausalRAG, and CTR for an example query:

Query event (28 Feb 2022): *"European natural gas prices surge 60% as Germany halts Nord Stream 2 pipeline certification amid Ukraine invasion."*

Vector Retrieval returns zero causally upstream events, retrieving instead a set of semantically related but non-causal energy market articles (storage levels, LNG imports) that pre-date the invasion without propagating toward it. CausalRAG retrieves two correct causal predecessors (NS2 halt, EU sanctions) but misranks the most proximate upstream event (Russia's gas transit cut on 24 February) below less relevant items. CTR correctly identifies four of five causal predecessors, ranking the Russia–Ukraine pipeline cut first and recovering the NATO signalling event—a cross-domain geopolitical trigger—as the fifth result. This qualitative comparison illustrates precisely the failure mode that motivates CTR: semantic retrieval conflates topical similarity with causal precedence, whereas directed causal edges with temporal decay preserve the asymmetric upstream-to-downstream signal.

7.7 Error Analysis

Despite its overall superiority, CTR fails on approximately 12% of crisis queries. We categorise failure modes and their approximate frequencies:

- **Black-swan events (51% of failures).** Genuinely unprecedented events—such as the initial COVID-19 lockdown announcements in March 2020—have no historical causal predecessors in $\mathcal{G}$. CTR degenerates to semantic retrieval for these queries because $W_{qi} \approx 0$ for all candidates, eliminating the causal signal entirely.
- **New entity emergence (28% of failures).** Entities that enter the graph only during the crisis period (e.g., newly sanctioned Russian banks, Ukrainian grain exporters) lack estimated causal edges, causing CTR to retrieve semantically similar but causally unrelated events.
- **Long causal chains (≥3 hops; 13% of failures).** CTR's one-hop retrieval objective does not explicitly model multi-hop causal chains. For queries where the causal predecessor is three or more hops upstream, retrieval precision drops to 0.61, approaching CausalRAG performance.
- **Ambiguous causal direction (8% of failures).** In some event pairs, transfer entropy is near-symmetric ($|\mathrm{TE}(X \to Y) - \mathrm{TE}(Y \to X)| < 0.05$), making the directional signal unreliable. These cases often correspond to mutually reinforcing feedback loops (e.g., inflation expectations and interest rate decisions), where causal direction is genuinely contested even among human experts.

These failure modes motivate the online learning and multi-hop retrieval extensions described in Section **??**.

7.8 Discussion

Taken together, the results establish three conclusions with broad implications for financial intelligence system design.

Causal direction is the single most valuable structural property. Across all experiments, the transition from undirected to directed causal edges consistently produces the largest performance gain—larger than adding temporal decay, transfer entropy, or hard-negative training individually. This has a practical implication: even a simple directed graph with approximate causal weights substantially outperforms a sophisticated undirected graph with exact correlation weights. System designers should prioritise causal directionality over correlation precision.

Temporal decay provides disproportionate robustness gains. While temporal decay contributes modestly to in-distribution performance (+1 NDCG point, Table 10), it reduces OOD degradation by 12.3 percentage points (from 22.1% to 9.8%, Table 8). This asymmetry—small in-distribution cost, large out-of-distribution benefit—makes learnable temporal decay an efficient design choice for any retrieval system deployed in non-stationary environments.

Hard-negative training bridges causal and semantic reasoning. The final gap between CTR and CausalRAG (7.4% NDCG improvement) is entirely attributable to hard-negative contrastive training, which forces the model to discriminate causal from semantic relevance. This suggests that even systems with identical graph structures can benefit substantially from more careful negative sampling strategies—a finding with immediate practical applicability to deployed retrieval systems.

8 Conclusion and Future Work

In this paper, we proposed **Causal Temporal Retrieval (CTR)**, a causal-aware financial event retrieval framework designed to improve retrieval robustness under temporal market distribution shifts. Unlike conventional financial retrieval systems that rely primarily on semantic similarity or correlation-based graph propagation, the proposed framework explicitly models directed causal relationships and temporal event asymmetry within a unified dynamic graph retrieval architecture.

The proposed method extends traditional vector retrieval by integrating semantic embeddings, temporal causal graph reasoning, and temporal decay regularization into a joint retrieval optimization framework. Specifically, CTR constructs directed causal edges using a hybrid causal discovery strategy based on **Granger causality** and **transfer entropy**, enabling the framework to capture both linear and nonlinear financial event propagation dynamics. Furthermore, a learnable temporal decay mechanism dynamically attenuates stale historical events while preserving influential upstream causal drivers, improving retrieval relevance under evolving market conditions.

The proposed causal-aware retrieval objective jointly optimizes semantic relevance, causal propagation strength, and temporal importance, allowing the framework to retrieve causally meaningful financial events instead of relying solely on semantic correlation. By incorporating directed temporal propagation into retrieval optimization, CTR substantially improves robustness during crisis-driven

Table 11: Top-5 retrieved events for an example query about European natural gas price surge (28 February 2022). Expert causal relevance label in brackets: [C] causally upstream, [S] semantically related but not causal, [N] not relevant.

Rank	Vector Retrieval	CausalRAG	CTR (Proposed)
1	EU energy minister discusses gas reserves (Jan 2022) **[S]**	Russia announces partial troop withdrawal (Feb 2022) **[S]**	Russia cuts gas transit via Ukraine pipeline (24 Feb 2022) **[C]**
2	UK gas prices hit record high (Dec 2021) **[S]**	Germany halts NS2 certification talks (Jan 2022) **[C]**	Germany halts NS2 certification talks (Jan 2022) **[C]**
3	LNG shipments to Europe increase (Jan 2022) **[S]**	EU sanctions target Russian energy sector (Feb 2022) **[C]**	EU sanctions target Russian energy sector (Feb 2022) **[C]**
4	Gas storage levels fall below 5-year average (Feb 2022) **[S]**	Gas storage levels fall below 5-year average (Feb 2022) **[S]**	Gazprom reduces eastward flows citing maintenance (22 Feb 2022) **[C]**
5	France considers nuclear energy expansion (Feb 2022) **[N]**	LNG shipments to Europe increase (Jan 2022) **[S]**	NATO member states signal support for Ukraine (24 Feb 2022) **[C]**
P@5	0/5 = 0.00	2/5 = 0.40	4/5 = 0.80

market transitions such as the COVID-19 pandemic, the Russia–Ukraine conflict, and inflation-driven macroeconomic regime shifts.

Experimental evaluation on heterogeneous financial event streams demonstrates that CTR consistently outperforms vector-based retrieval, graph-based retrieval, and causal retrieval baselines across multiple evaluation metrics including Precision@K, Recall@K, MRR, NDCG, AUC, and robustness degradation percentage. Compared with semantic-only retrieval systems, the proposed framework exhibits significantly lower retrieval degradation during out-of-distribution crisis periods, indicating that causal temporal graph structure provides a stronger inductive bias for financial event intelligence under rapidly evolving market conditions.

The ablation study further confirms that each component contributes meaningfully to retrieval performance and robustness. Directed causal propagation produces substantial improvement over undirected graph retrieval, while temporal decay regularization improves temporal generalization by suppressing outdated event influence. Additionally, transfer entropy contributes important non-linear causal information beyond standard Granger causality, highlighting the importance of modeling complex financial dependencies in dynamic market environments.

A theoretical robustness proposition further suggests that temporal causal dependencies remain more stable than correlation-based co-occurrence statistics under moderate distribution shifts, providing a principled explanation for the improved generalization behavior observed experimentally.

Overall, the findings demonstrate that integrating causal reasoning with temporal graph retrieval represents a promising direction for next-generation financial intelligence systems capable of operating under rapidly evolving market conditions.

8.1 Future Work

Several promising research directions remain for extending the proposed CTR framework.

- **Multimodal Financial Intelligence**: The current framework primarily focuses on textual and numerical financial events. Future work may incorporate satellite imagery, shipping trajectories, supply-chain sensor data, and alternative financial signals to enable multimodal causal event reasoning.
- **Temporal Graph Neural Networks**: Future extensions may integrate advanced temporal graph neural network architectures such as TGAT, TGN, or temporal graph transformers to improve long-range causal propagation learning.
- **Online Causal Learning**: The current framework performs periodic graph updates. Future systems may support fully online causal graph adaptation under streaming financial event environments using continual learning strategies.
- **Large-Scale Financial Retrieval**: Scaling CTR to billion-scale financial event graphs will require distributed graph retrieval, approximate nearest-neighbor indexing, and memory-efficient temporal graph optimization.
- **Cross-Market Causal Transfer**: Future work may investigate whether causal propagation learned in one financial market can generalize across international markets, asset classes, or macroeconomic environments.
- **LLM-Augmented Financial Reasoning**: Large language models may be integrated with causal retrieval pipelines to provide interpretable financial event reasoning and multi-hop market explanation generation.
- **Theoretical Generalization Guarantees**: Stronger theoretical analysis of causal temporal retrieval under distribution shifts and dynamic graph evolution remains an important open research problem.

In conclusion, this work demonstrates that financial event retrieval systems should move beyond semantic similarity and correlation-based retrieval toward causal temporal graph reasoning. The proposed CTR framework establishes a foundation for robust causal-aware financial intelligence capable of adapting to rapidly changing global market dynamics.

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